

1

2

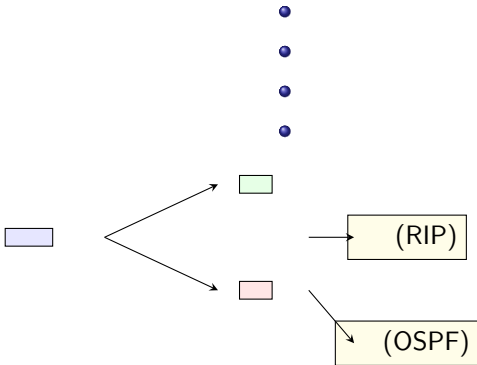
RIP

3

OSPF

4

BGP



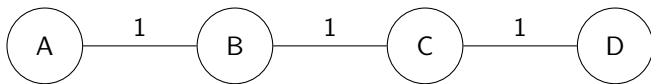
Bellman-Ford

$$d_x(y) = \min\{c(x, v) + d_v(y)\}$$

- $d_x(y)$ $x \rightarrow y$
- $c(x, v)$ $x \rightarrow v$
- $d_v(y)$ $v \rightarrow y$

RIP

- $\leq 15 \cdot 16$
- 30
- UDP 520
-



A

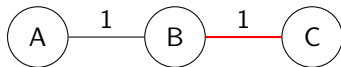
- A: 0,
- B: 1, B

B

- C: 2, B
- D: 3, B

- 1 v
- 2 y
- 3 $= c(x, v) + d_v(y)$
- 4 $<$

RIP



- C A C → B → A 2
- C → B B A C B A C
- B C A 2 ⇒ C A 3
- C B A 3 ⇒ B 4
- 16 16 =

Split Horizon

— OSPF

LS

Dijkstra

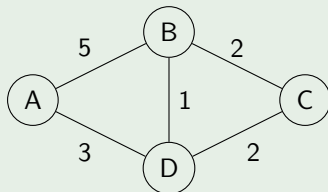
OSPF

- 1
- 2 LSP
- 3 LSP
- 4
- 5 Dijkstra

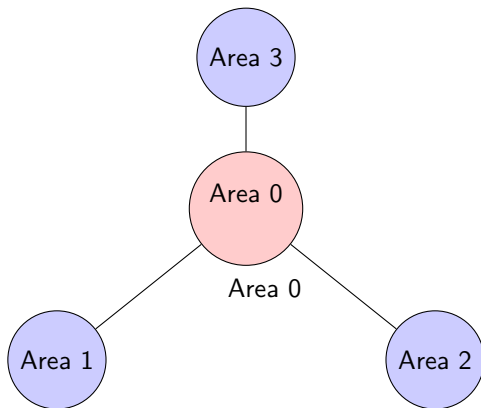
-
-
- IP 89
-
-

Dijkstra

- 1 0 ∞
- 2 u
- 3 u
- 4 u $d(v) = \min\{d(v), d(u) + c(u, v)\}$
- 5 2-4



OSPF —



- **Area 0**
- ABR
- LSA

VS

	RIP	OSPF
	Bellman-Ford 15 (30s)	Dijkstra

- RIP
- OSPF

	RIP	OSPF	BGP
	(IGP)	(IGP)	(EGP)
	UDP 520	IP 89	TCP 179

- IGP AS RIP OSPF
- EGP AS BGP